TurboQuant v1.5.2

Work Log

2026-04-07

PPL 21%→8%, Attention Sharpening + V Rotation Bugfix

MMSE-derived α

attn_rot_v bug found

Split α for TBQ3/TBQP3

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Session Overview

- `Morning` — PPL benchmark investigation — discovered 21% gap vs f16 at ctx=2048
- `Morning` — Code audit — found 1.15x V heavy-tail compensation was empirical hack
- `Afternoon` — Per-block norm implementation (global norm → per-block for TBQ3 D=512)
- `Afternoon` — Removed 1.15x V compensation
- `Afternoon` — Sub-block scale experiment (32-element groups) — proved WHT already handles variance uniformization
- `Evening` — Breakthrough — Attention Sharpening $\alpha = 1 + 1/(2 \times \mathrm{SQNR})$ derived from MMSE theory
- `Evening` — V rotation bug discovered (attn_rot_v=1 but IWHT has no inverse rotation)
- `Night` — Split α for TBQ3 (1.016) vs TBQP3 (1.036) — different noise profiles
- `Night` — Final PPL 454.7 (1.08x vs f16), math bench 19.1/35 vs f16 20.1/35

PPL Gap Investigation Problem

SYMPTOMS

- v1.5.1: PPL 509.9 (vs f16 419.8 = **1.21x**)
- q4_0 only 1.03x gap → TBQ3's 1.21x is disproportionate
- Why is 3-bit so much worse than 4-bit?

ROOT CAUSE

- Not bit count, but **scale factor count**
- q4_0: 16 scales per 512 dims
- TBQ3: only **2 scales** per 512 dims
- Fewer scales → accumulated quantization noise

Key insight: TBQ3's problem is not 3-bit precision itself, but too few scale factors per block causing softmax distribution flattening. Instead of adding more scales, a different approach is needed.

Sub-block Scale Experiment Failed

EXPERIMENT DESIGN

- Added 8 sub-scales to `block_tbq3_0` struct
- 32-element groups with independent scales
- Goal: finer-grained scaling to improve PPL

RESULTS

- PPL **worsened**: 514.6 vs baseline 509.9
- Sub-scales themselves add quantization noise
- WHT already performs variance uniformization

Lesson: WHT (Walsh-Hadamard Transform) already uniformizes coefficient variance. Sub-scales are redundant and only add noise. → Pivoted to **attention correction** approach.

Attention Sharpening Breakthrough

THEORETICAL BASIS

- 3-bit quantization noise flattens softmax distribution
- Derived from MMSE theory: `α = 1 + 1/(2×SQNR)`
- $SQNR_{3bit} \approx 13.8 \rightarrow \alpha \approx 1.036$

ADVANTAGES

- **Zero memory cost** — one scalar multiply
- **Zero compute cost** — single multiply before softmax
- Theoretically grounded correction (not empirical hack)

Result: PPL 509.9 → **455.0** (1.21x → **1.08x**). 62% reduction in PPL gap with zero additional memory or compute cost.

V Rotation Bug

Bugfix

PROBLEM

- `attn_rot` applied to both K and V
- K: safe — Q gets same rotation, cancels in dot product
  `(R·Q)·(R·K) = Q·K`
- V: **BUG** — IWHT decode has no inverse rotation
- V output remains rotated → corrupted results

FIX

- Set `attn_rot_v = false`
- Only K gets rotation, V keeps original values
- IWHT correctly recovers original V
- PPL: 455.0 → **448.1** (1.08x → 1.07x)

Principle: K rotation cancels via $R^T R = I$ in the Q·K dot product, but V is directly output after weighted sum — any remaining rotation distorts the result.

Split α for TBQ3 / TBQP3 Optimization

TBQP3 (2-BIT + QJL)

- QJL projection adds extra noise to attention scores
- Needs stronger sharpening
- **α = 1.036**

TBQ3 (3-BIT)

- Cleaner noise profile
- Lighter sharpening sufficient
- **α = 1.016**

Result: Split α vs single α improved tbqp3/tbq3 PPL: 473.7 → **454.7**. Each quantization method gets correction matched to its noise characteristics.

Per-block Norm Improvement

TBQ3 (3-BIT)

- Per-block norm applied
- Each 256-half normalized independently
- Captures per-block variance differences accurately

TBQP3 (2-BIT + QJL)

- Global norm retained
- QJL residual uses cross-block 512-WHT
- Requires information sharing across blocks

Design principle: TBQ3 uses independent 256-blocks → per-block norm is natural. TBQP3 uses full 512-WHT → global norm is the correct choice.

tbq4_0 OOB Bug Fix

Bugfix

PROBLEM

- `vec_dot_fattn_vec_KQ_tbq4_0`
- At D=512, only read `K_tbq[0]`
- Remaining 256 dims unprocessed → incorrect results

FIX

- Applied `n_blocks = D/256` loop
- Iterates over all blocks for complete dot product
- Works correctly for both D=256 and D=512

Final Results v1.5.2

METRIC	F16	TBQP3/TBQ3	RATIO
PPL (ctx=2048)	419.8	**454.7**	**1.08x**
Math Bench	20.1/35	19.1/35	95%
Math Peak	21	**23**	**>f16**
KV Memory	5,420 MiB	**1,290 MiB**	**4.2x compression**

1.08x
PPL (v1.5.1: 1.21x)

4.2x
KV Cache Compression

0
Extra Memory Cost

62%
PPL Gap Reduction

v1.5.2 Summary

- Attention Sharpening: MMSE theory-derived α, zero memory/compute cost
- V Rotation bug found and fixed: IWHT inverse lacked rotation reversal
- Split α for TBQ3/TBQP3: optimal correction per noise profile
- Per-block norm: TBQ3 block-independent, TBQP3 global retained
- Sub-block scale experiment: proved WHT variance uniformization (negative result has value)
- tbq4_0 OOB bug fix: D=512 multi-block loop

v1.5.1 → v1.5.2: PPL 509.9 → 454.7 (1.21x → 1.08x). A theoretically grounded single-scalar correction dramatically improved quantization quality. Achieved within 8% of f16 PPL at 4.2x compression.